 in goblet cells which accompanies the inflammatory process is believed to be caused by increased expression of the transcription factors: Hath1 (a basic helix-loop-helix (bHLH) transcription factor) and Klupper like factor 4 (KLF4), which are essential for goblet cell differentiation [15]. This inflammatory process is believed to drive the increase in goblet cell differentiation, which is observed in Crohn's disease patients [15]. However, it is important to note that this increase in goblet cells is not found in all Crohn's disease patients, or throughout the whole gastrointestinal tract [16]. Maintenance of intestinal homeostasis is regulated by epithelial barrier integrity [17]. Diffusion of the luminal materials into lamina propria promotes a local inflammatory response causing release of proinflammatory cytokines, release of MMPs, and epithelial degradation and inflammation [18]. Expression of pro-inflammatory cytokines: IL-1α, IL-1β, TNFα, and IL-15 were increased in IL-1rn-/- mice, and these results are in agreement with those obtained from patients with Crohn's disease [19]. IL-1α and IL-1β are also expressed in human intestinal epithelial cells [20]. IL-1α is released when these human intestinal cells are damaged or destroyed, and IL-1α was commonly detected in the epithelium of IBD, released from necrotic intestinal epithelial cells and plays a crucial role in intestinal inflammation by inducing human intestinal fibroblasts to produce IL-6 and IL-8 [21]. The high concentrations of IL-1β in the intestine of patients with Crohn's disease are mainly attributed to local mononuclear cell infiltration [22], which were also observed in this model. Increased production of IL-1β and TNFα in inflamed Crohn's disease mucosa induces the synthesis of IL-8, which is an effective neutrophil chemoattractant [23]. These findings agree with previous work, which showed increased numbers of TNFα positive cells in the lamina propria and submucosa of patients with Crohn's disease. Furthermore, TNFα produced by subepithelial macrophages has been shown to contribute to decreased epithelial integrity [24]. TNFα acts as an important mediator of inflammation through MAPK and NF-κB activation, increasing cell proliferation and altering epithelial barrier permeability [25]. The production of TNFα is also high in cultured mucosal mononuclear cells from Crohn's disease patients [26]. However, in contrast a previous study observed no differences in TNFα mRNA expression between control and IBD mucosal biopsies [27]. IL-1RI is expressed on many cell types including T cells, fibroblasts, and intestinal epithelial cells. Despite the increased expression of IL-1β in the IL-1rn-/- mice, the expression of IL-1R1 was reduced; this could suggest a negative feedback loop decreasing expression of IL-1RI in the presence of uncontrolled IL-1 production. This agrees with a previous study, which showed IL-1β and TNFα treatment downregulated the expression of mRNA IL-1RI in rat intestinal epithelial cells (IEC-6) [28]. Similarly, IL-1β treatment was also shown to downregulate IL-1RI protein in retinal endothelial cells (TR-iBRB2) [29]. This is contrary to previous study, in which an increase in IL- 1β induced chronic intestinal inflammation in mice and an increase in IL-17A-secreting innate lymphoid cells which express high levels of IL-1R1 [30]. These differences may be a result of the different levels of inflammation seen in these models. Here, increased expression of IL-15 was seen in IL-1rn-/- mice, this is in agreement with Liu et al (2000), who showed increased expression of IL-15 by macrophages in the inflamed ileum of patients with Crohn's disease and that IL-15 increased local T cell activation and induced proinflammatory cytokine production by T cells and macrophages. The presence of polymorphonuclear cells and mononuclear cells play an important role in the augmentation of inflammation and tissue damage in IBD [31]. Increased infiltration of these CD11b and CD68 positive cells into the lamina propria were observed in IL-1rn-/- mice compared with WT mice. These findings are in agreement with previous findings which showed increased numbers of these cells in the inflamed mucosa of IBD patients [32]. Infiltration of immune cells such as macrophages and lymphocytes in the lamina propria is an important aspect of IBD, especially Crohn's disease, even in the lack of noticeable morphological, clinical, and endoscopic indication of inflammation. This is thought to be due to an increased demand of macrophages within the inflamed intestine [33]. Functional assessment of this increased inflammatory response such as calprotectin measurement in stool samples would have been a useful assessment, however unfortunately these samples were not available in this study. MMPs and ADAMTS1 expression were not significantly altered in IL-1rn-/- mice. MMPs have been demonstrated to be induced in several pathological conditions of IBD and play a key role in regulating the pathophysiology of IBD [34]. The main action of MMPs is to degrade extracellular matrix proteins. Active MMP2 and MMP9 play an important role in cell migration and cytokine stimulation. Indeed neutrophil infiltration has been shown to be induced by upregulation of MMP9 in murine inflamed intestine [35]. Furthermore, stromal cells play an important role in intestinal inflammation and the pathogenesis of IBD. These cells secrete MMP2 and MMP9 [36]; and MMP9 is produced by human colonic epithelium during IBD [37]. In addition, infiltrating macrophages and neutrophils are the main source of MMP9 in human IBD [38]. Overexpression of MMP9 has been shown to cause a reduction in the differentiation of progenitor cells to goblet cells and consequently decreased MUC2 expression, which leads to decreases in the protective mucin barrier in colonic epithelium [35]. The maintenance of epithelial barrier function and control of paracellular permeability are regulated by tight junction proteins [39]. Epithelial disruption and decrease in tight junctions that lead to increased intestinal permeability are the common features of Crohn's disease [40]. Decreased expression of ZO-1 and loss of E-cadherin